

Project Report: SaaS Subscription Platform - Full Stack Web Developer

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Family:	Full Stack Web Developer
Level:	Advanced
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Multi-tenant SaaS with Stripe billing, team invites, admin panel, and observability.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Advanced]

Chapter focus: Project Overview** *(Level: Advanced)

SaaS Subscription Platform** is a advanced-level end-to-end project for Full Stack Web Developer. **Duration:** 6-8 weeks. **Multi-tenant SaaS with Stripe billing, team invites, admin panel, and observability.** **Tools:** React, Node.js, Stripe, PostgreSQL.

Code Reference:

Objective: Stripe subscription checkout
Objective: Multi-tenant data isolation
Objective: Team invite flow
Objective: Admin analytics panel

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Full Stack Web Developer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Full Stack Web Developer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Billing [Advanced]

Chapter focus: Phase: Billing** *(Level: Advanced)

Stripe integration. Complete these tasks: Checkout session; Webhooks. Deliverable for this phase: Test subscription flow.

Code Reference:

- [] Checkout session
- [] Webhooks

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Billing is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Full Stack Web Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Test subscription flow to the teacher.

Chapter 3: Phase: Tenants [Advanced]

Chapter focus: Phase: Tenants** *(Level: Advanced)

Org-scoped data. Complete these tasks: Middleware tenant ID; Row isolation. Deliverable for this phase: Multi-tenant tests.

Code Reference:

- [] Middleware tenant ID
- [] Row isolation

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Tenants is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Full Stack Web Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Multi-tenant tests to the teacher.

Chapter 4: Phase: Teams [Advanced]**Chapter focus: Phase: Teams** *(Level: Advanced)**

Invite and roles. Complete these tasks: Email invite; RBAC. Deliverable for this phase: Team management UI.

Code Reference:

- ☐ Email invite
- ☐ RBAC

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Teams is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Full Stack Web Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Team management UI to the teacher.

Chapter 5: Phase: Ops [Advanced]**Chapter focus: Phase: Ops** *(Level: Advanced)**

Logging and monitoring. Complete these tasks: Structured logs; Health checks. Deliverable for this phase: Ops runbook.

Code Reference:

- ☐ Structured logs
- ☐ Health checks

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Ops is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Full Stack Web Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Ops runbook to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Advanced]

Chapter focus: Final Deliverables and Review *(Level: Advanced)**

Submit all final deliverables: Live SaaS app; Stripe test receipts; Ops runbook; Architecture diagram. Skills demonstrated: SaaS, Billing, Multi-tenant. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Live SaaS app
Deliverable: Stripe test receipts
Deliverable: Ops runbook
Deliverable: Architecture diagram
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Live SaaS app and passes the rubric.